HHH-C-002167A(Army-GL)
August 6, 1985
SUPERSEDING
HHH-C-002167
June 24, 1983

INTERIM FEDERAL SPECIFICATION

CORN, SWEET, CREAM STYLE, THERMOSTABILIZED, TRAY PACK

This interim Federal specification was developed by U.S. Army Natick Research and Development Center, Natick, MA 01760-5014, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE

1.1 Scope. This specification covers cream-style, sweet corn, thermostabilized in tray pack cans for use by the Department of Defense as a component of operational rations.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

TT-C-495 - Coatings, Exterior, for Tinned Food Cans
PPP-B-636 - Boxes, Shipping, Fiberboard

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification, other Federal specifications, and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY;

FSC 8915

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:44Z
Check the source to verify that this is the current version before use.

Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Houston, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.

(Federal Government activities may obtain copies of Federal standardization documents, and the Index of Federal Specifications, Standards and Commercial Item Descriptions from established distribution points in their agencies.)

Military Specifications:

MIL-L-1497 - Labeling of Metal Cans for Subsistence Items
MIL-L-35078 - Loads, Unit: Preparation of Nonperishable Subsistence Items; General Specification for

Military Standards:

MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-900 - Bacterial Standards for Starches, Flours, Cereals, Alimentary Pastes, Dry Milks and Sugars Used in the Preparation of Thermostabilized Foods for the Armed Forces

(Copies of Military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Laws and Regulations:

U.S. Department of Health and Human Services

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder (21 CFR parts 1-199)

(Applications for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

U.S. Department of Agriculture (USDA)

U.S. Standards for Conditions of Food Containers

(Applications for copies should be addressed to the Director, Market Research and Development Division, Agricultural Marketing Service (AMS), U.S. Department of Agriculture, Washington, DC 20250.)

U.S. Standards for Grades of Canned Cream Style Corn

U.S. Standards for Grades of Frozen Whole Kernel (or Whole Grain) Corn

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:44Z
Check the source to verify that this is the current version before use.

(Application for copies should be addressed to the Chief, Processed Products Branch, Information Division, Agricultural Marketing Service, U.S. Department of Agriculture, Washington, DC 20250.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

Association of Official Analytical Chemists (AOAC)

Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, 1111 North 19th Street, Suite 210, Arlington, VA 22209.)

Environmental Protection Agency (EPA)

National Interim Primary Drinking Water Regulations

(Application for copies should be addressed to the Office of Drinking Water WH550, Environmental Protection Agency, Washington, DC 20460.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.3, 6.1 and 6.3).

3.2 Ingredients. All ingredients shall be clean, sound, wholesome, and free from foreign material, evidence of rodent and insect infestation, extraneous material, off-odors, off-flavors, and off-colors.

3.2.1 Corn. The corn shall be fresh, succulent, mature and of a single yellow variety suitable for canning. Alternatively, the corn may be frozen, yellow, U.S. Grade A of the U.S. Standards for Grades of Frozen Whole Kernel (or Whole Grain) Corn. The moisture content of the fresh or frozen corn shall be not less than 68 percent nor more than 71 percent as determined in accordance with the methods described in the Official Methods of Analysis of the Association of Official Analytical Chemists, Chapter: Vegetable Products Processed, Paragraph: Solids(Total) in Canned Vegetables, and the corn shall be of the latest season's crop or pack.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:44Z
Check the source to verify that this is the current version before use.

3.2.2 Starch, modified, high opacity. The high opacity starch shall be white, odorless, finely pulverized, modified maize food starch for use in canned foods. The modified starch shall demonstrate initial viscosity development in the temperature range 140° to 170°F and a typical viscosity (be fully hydrated) at common retort temperatures. The starch shall resist breakdown at low pH, under shear stress and conditions of cold storage. The starch shall be bland with essentially no cereal or starch taste and shall comply with MIL-STD-900.

3.2.3 Salt. Salt shall be non-iodized, white, refined, sodium chloride, with or without anti-caking agents.

3.2.4 Sugar. Sugar shall be white, refined, sucrose, granulated cane or beet or a combination of both and shall comply with MIL-STD-900.

3.2.5 Water. Water used for formulation and washing shall conform to the National Interim Primary Drinking Water Regulations.

3.3 Preparation and processing.

3.3.1 Corn preparation.

3.3.1.1 Fresh corn preparation. The fresh corn ears shall be husked, silked, cooled, sorted and washed. The corn kernels shall be cut from the ears and the ears scraped to remove the remaining portion of the kernels from the cob. The cutter knives shall be adjusted to approximately cut the kernels in half.

3.3.1.2 Frozen corn preparation. The frozen corn shall be thawed prior to grinding.

3.3.2 Creamed corn preparation. The required percentage of corn kernels (see 3.3.3) together with the corn scrapings shall be mechanically ground through a comminuter to form a smooth, semi-liquid creamy portion.

3.3.3 Cream style corn preparation. The cream style corn shall be prepared from the following ingredients:

Ingredients	Percent
Corn kernels (not ground)	39.0
Creamed corn	22.0
Water 1/	32.0
Sugar	4.3
Starch, modified 2/	1.7
Salt	1.0

1/ The percent of water shall be adjusted appropriately depending on the maturity of the corn utilized to meet the consistency requirement of the finished product (see 3.6).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:44Z
Check the source to verify that this is the current version before use.

2/ The percent starch shall be adjusted appropriately to meet the consistency requirement of the finished product (see 3.6).

a. The whole kernel corn, creamed corn, sugar, salt and approximately 75 percent of the water shall be heated to 180° to 190°F with continuous agitation.

b. A thin slurry shall be prepared from the remaining water and the modified starch using a high speed mixer.

c. The starch slurry shall be added to the corn mixture and the entire contents heated to 190°F and held for approximately 5 minutes.

d. The volume of the final mixture shall be adjusted with water to compensate for evaporation loss during heating.

e. The product shall be filled into the tray pack cans within 2 hours after preparation.

3.4 Tray pack filling and sealing. Each tray pack can (see 5.1.1) shall be filled with cream style corn such as to conform to finished product requirements and to the following requirements:

a. Each tray pack can shall be filled with sufficient corn to meet the finished product net weight requirements (see 3.6).

b. The temperature of the cream style corn at the time of filling shall be not less than 180°F.

c. There shall be a headspace of at least 1/4 inch after filling.

d. The filled and sealed tray pack cans shall be in the retort process within 2 hour after sealing.

e. The cans shall be hermetically sealed with a minimum of 10 inches of mercury (Hg) immediately after filling.

3.5 Tray pack thermoprocessing. The filled and sealed tray pack cans shall be thermoprocessed by retorting at 250°F or below until a sterilization value (F_o) of not less than 5.0 has been achieved. The cans shall be cooled under pressure to a can center temperature of 95°F or less prior to packing.

3.6 Finished product requirements. The finished product shall comply with the following requirements:

a. There shall be no foreign material such as but not restricted to glass, paint, metal, dirt, insects, insect parts, hair, wood, or paper.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:44Z
Check the source to verify that this is the current version before use.

b. There shall be no foreign odors or flavors such as but not restricted to rancid, scorched, burnt, stale, sour or moldy.
c. There shall be no color foreign to the product.
d. The average net weight shall be not less than 106 ounces.
e. No individual can shall have a net weight of less than 104 ounces.
f. The finished product shall be U.S. Grade A in accordance with the U.S. Standards for Grades of Canned Cream Style Corn.
g. No individual can shall have a salt content of less than 0.5 percent or greater than 1.5 percent.

3.6.1 Palatability. The finished product shall be equal to or better than the approved preproduction sample (see 6.1) in palatability and overall appearance.

3.7 Plant qualification. The product shall be prepared, processed, and packaged in establishments meeting the requirements of Title 21, Code of Federal Regulations, Part 110, Current Good Manufacturing Practice in Manufacturing, Processing, Packaging or Holding of Human Foods, and the plant sanitation requirements of the appropriate Government Inspection agency.

3.8 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act, and regulations promulgated thereunder.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a product complying with all requirements of this specification. The contractor shall assure product compliance prior to submitting the product to the USDA for any inspection.

4.2 Inspection and certification. Product acceptability shall be determined by the USDA. The USDA will determine the degree of supervision necessary to assure compliance with the requirements of this specification.

4.3 First article inspection. When a first article is required (see 6.1), it shall be inspected in accordance with the quality assurance provisions of this specification and evaluated for overall appearance and palatability. Any failure to conform to the quality assurance provisions of this document or any appearance or palatability failure shall be cause for rejection of the first article.

4.4 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or other applicable purchase document.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:44Z
Check the source to verify that this is the current version before use.

4.4.1 Ingredient and component examination. Conformance of ingredients and components to identity, condition and other requirements specified in 3.2 shall be certified by the ingredient supplier or ingredient manufacturer, or compliance be evident by examination of pertinent labels, markings, U.S. Grade certificates, certificates of analysis, or other such valid documents acceptable to the inspection agency. In addition, prior to use each ingredient shall be examined organoleptically, as necessary to determine conformance to the condition requirements. Any nonconformance to an identity, condition, or other requirement shall be cause for rejection of the ingredient or component lot or of any involved product.

4.5 In-process examination. In-process examination shall be performed to determine conformance to the preparation, processing, can interior coating, filling, sealing, and packing requirements. Any nonconformance revealed by actual examination or by review of records of time, temperature, and formulation or of other valid documents shall be cause for rejection of the involved product.

4.6 Tray pack inspection. The inspection lot shall include only tray packs produced in one workshift. The USDA reserves the right to separate the inspection lot into smaller inspection lots.

4.6.1 Net weight inspection. Randomly select 30 filled and sealed tray pack cans from the inspection lot and weigh separately. Subtract the average tare weight (determined by randomly selecting and weighing 30 of the empty tray pack cans and lids used in preparing the product and dividing the total weight by 30) from the weight of each tray pack in the sample. The results shall be reported to the nearest 1 ounce. If the average net weight is less than 106 ounces or if the net weight of any individual can is less than 104 ounces, the lot shall be rejected.

4.6.2 Product inspection. The sample size shall be as indicated by the double sampling plan specified in table I.. The tray pack cans shall be selected at random from the lot. The tray pack cans shall be opened and inspected for the defect listed in table II.

TABLE I. Double sampling plan for product inspection. 1/

Lot size (cans)	Sample size (No. of cans)	Cumulative sample	Acceptance number	Rejection number
0-3200	8		0	2
	8	16	1	2
3201-35000	13		0	3
	13	26	3	4

1/ a. If no defects are found in the first sample, the lot shall be accepted.

Source: https://assist.dla.mil -- Downloaded: 202610-26T12:44Z
Check the source to verify that this is the current version before use.

b. If the number of defects found in the first sample equals or exceeds the rejection number, the lot shall be rejected.

c. If the number of defects found in the first sample exceeds the acceptance number but is less than the rejection number, the second sample shall be inspected. Defects found in the first and second samples shall be combined and if the number of defects in the cumulative sample equals or exceeds the rejection number, the lot shall be rejected.

TABLE II. Product inspection 1/ 2/

Category	Defect
Major	
101	Not grade specified. 3/

1/ The presence of any foreign material (e.g. glass, dirt, insect parts, hair, wood, metal, etc), foreign odor or flavor (e.g. burnt, scorched, moldy, rancid, sour, stale, etc.) or foreign color shall be cause for rejection of the lot. (To be determined by an AMS agent.)

2/ Product not equal to or better than the approved preproduction sample (see 6.1) in palatability and overall appearance shall be cause for rejection of the lot. (This comparison shall be performed only when deemed necessary by an AMS agent.)

3/ The product shall be inspected in accordance with the provisions of the U.S. Standards for Grades of Canned Cream Style Corn and Regulations Governing Inspection and Certification of Processed Fruits and Vegetables, Processed Products Thereof, and certain other Processed Food Products.

4.6.3 Salt content testing. Nine tray packs shall be selected from the lot and distributed as follows:

-three for laboratory analysis
-three for submission to the contractor
-three for retention by the AMS agent as a reserve sample.

The three tray packs for laboratory analysis shall be individually tested for salt content in accordance with the Official Methods of Analysis of the Association of Official Analytical Chemists, except that preparation of the sample shall be as follows. The cans shall be opened and the entire contents of each can shall be separately blended in a Waring blender or equivalent. The lot shall be rejected if:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:44Z
Check the source to verify that this is the current version before use.

The salt content in any individual tray pack is greater than 1.5 percent or less than 0.5 percent. Results shall be reported to the nearest 0.1 percent.

Analysis of reserve samples at the request of the contractor shall not be permitted unless the original laboratory analysis indicated that the involved lot will be rejected because of noncompliance with the salt content requirement. When the reserve sample is analyzed, the analyses for salt shall be made and will be considered final. Unused reserve samples shall be returned to the contractor for inclusion in subsequent lots.

4.7 Can condition examination. Examination of filled and sealed tray pack cans shall be in accordance with the United States Standards for Condition of Food Containers, except that inspection for labeling shall be in accordance with MIL-L-1497 (see 5.4).

4.8 Shipping container examination. Shipping containers shall be examined for defects in assembly, closure, and reinforcement (when applicable), in accordance with the appendix of PPP-B-636. In addition, the following defects shall be classified as follows: Major: (1) National stock number, item description, contract number, or date of pack markings missing, incorrect, or illegible. (2) Reinforcement with nonmetallic strapping or tape is not used. Minor: (1) Other required markings missing, incorrect, or illegible; (2) Interior packing with fiberboard liner or pads not as specified; (3) Height of box liner less than inside depth dimension of box by more than 1/8 inch or extends above top edge of box; (4) Length or width of pads less than box dimensions by more than 1/8 inch; (5) Arrangement or number of cans not as specified.

4.9 Unit load inspection. Inspection of unit loads shall be in accordance with the quality assurance provisions of MIL-L-35078.

5. PREPARATION FOR DELIVERY

5.1 Packaging. The product shall be packaged in accordance with level A or C as specified (see 6.1).

5.1.1 Level A. A size 1001 by 1206 by 200 tray pack can shall be filled with 106 ounces of food product, sealed, and thermoprocessed as specified in 3.4 and 3.5. The base plate of the can shall be electrolytic chromium-coated steel and shall be of sufficient temper to protect the product during shipment and storage. The base plate weight of the body and end of each can shall be 90 pounds per base box. The entire inside area of the can shall be coated. Scratches or fractures shall not penetrate through the interior can coating. The interior can coating shall not peel or blister when in contact with the product. The interior can coating shall neither affect nor be affected by the packaged product. A certificate of conformance shall be furnished to certify compliance with the interior can coating requirements. The can shall be coated overall on the outside with a coating conforming to type I of TT-C-495.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:44Z
Check the source to verify that this is the current version before use.

5.1.2 Level C. The product shall be packaged as specified in 5.1.1, except that the cans with commercial exterior coating will be acceptable.

5.2 Packing. The product shall be packed in accordance with level A, B or C as specified (see 6.1).

5.2.1 Level A packing. Four cans of product, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard box, constructed, and closed in accordance with style RSC-L, grade V2s of PPP-B-636. The cans shall be packed flat, four in depth within the box. The inside of each box shall be provided with a box liner and five fiberboard pads. The height of the box liner shall be equal to the full inside depth of the box (+0 inch, -1/8 inch). The pads shall be not less than 1/8 inch of the full length and width dimensions of the box and shall be fabricated of grade V3c fiberboard. The pads shall be placed between the cans and on the top and bottom of the stacked cans. Each fiberboard box shall be reinforced with nonmetallic strapping or pressure-sensitive adhesive filament reinforced tape in accordance with the appendix of PPP-B-636. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.1). Strapping shall be limited to nonmetallic strapping, except for type II. class F loads.

5.2.2 Level B packing. Four cans of product, packaged as specified in 5.1, shall be packed as specified in 5.2.1, except the box shall be constructed of grade V3c, V3s, or V4s fiberboard.

5.2.3 Level C packing. Four cans of product, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard box, constructed and closed in accordance with style RSC-L, class domestic, grade 275 of PPP-B-636. The cans shall be packed flat, four in depth within the box. The inside of each box shall be provided with a box liner and five fiberboard pads. The height of the box liner shall be equal to the full inside depth of the box (+0 inch, -1/8 inch). The pads shall be placed between the cans and on the top and bottom of the stacked cans. The pad dimensions shall be not less than 1/8 inch of the full length and width dimensions of the box and shall be fabricated of the same material as the box.

5.3 Unit loading. When specified (see 6.1), the product, packed as specified in 5.2.2 and 5.2.3, shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified. When unit loads are strapped, the strapping shall be limited to nonmetallic strapping.

5.4 Labeling. Each tray pack can shall be labeled in accordance with MIL-L-1497. In addition, the name of product shall be marked, stamping is permitted, on one 1001 by 200 side of the can. The labeling shall be legible when examined in accordance with 4.7 after submerging the can in boiling water for 30 minutes. Paper labels are not permitted. Cans shall show the following statements:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:44Z
Check the source to verify that this is the current version before use.

TO HEAT

a. To heat in water: Submerge unopened can in boiling water. Simmer gently 25 minutes. Avoid overheating to the point where internal pressure is evident (can shows evidence of bulging). Remove from water and open lid for serving. Caution: Use care when opening as pressure may have been generated within the can.

b. To heat in oven: Either punch several holes in lid of can or open can in usual manner leaving the loose lid in place to prevent moisture loss. Place in a 350°F oven for approximately 35 to 40 minutes until contents reach an internal temperature of 165°F.

WARNING: Do not place closed can in oven, as heating will increase internal pressure which may cause the can to burst.

YIELD

Approximately 25 servings of 1/2 cup each.

5.5 Marking.

5.5.1 Shipping containers. In addition to any special marking required by the contract, shipping containers shall be marked in accordance with MIL-STD-129.

5.5.2 Unit loads. Unit loads shall be marked in accordance with MIL-L-35078. In addition, the following precautionary markings in capital letters larger than other markings shall be included:

CAUTION: DO NOT STACK PALLETS IN TRANSIT OR MORE THAN TWO HIGH IN STORAGE.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. When a first article is required (see 3.1, 4.3, and 6.3).

Source: https://assist.dla.mil -- Downloaded: 202610-26T12:44Z
Check the source to verify that this is the current version before use.

c. Provisions for approved preproduction samples (see 3.6.1 and 6.3).
d. Level of packaging and packing required (see 5.1 and 5.2).
e. Type and class of unit load when unit loading is required (see 5.2.1 and 5.3).

6.2 Appropriate level of pack. Based on the conditions known or expected to be encountered during shipment, handling and storage of the specific item being procured, the procuring activity should select the appropriate level of pack in accordance with the criteria established in AR 700-15/NAVSUPINST 4030.28/AFR 71-6/MCO 4030.33A/DLAR 4145.7.

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

Preparing activity:

Army - GL

Project No. 8915-A788

 U.S. GOVERNMENT PRINTING OFFICE: 1985 - 505-038/23743

Source: https://assist.dla.mil -- Downloaded: 202610-26T12:44Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
HHH-C-002167A(Army-GL)	Corn, Sweet, Creamed Style, Thermostabilized, Tray Pack

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

NATICK OP, 20 Nov 84

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:44Z
Check the source to verify that this is the current version before use.